INTEGER PARAMETER SHIFTS IN ONE-VARIABLE RESTRICTIONS OF LAURICELLA'S F_C

NAKANO RYUNOSUKE

Abstract. We construct rational functions of the coefficient index for one-variable restrictions of Lauricella's function F_C at arguments related by cube roots of unity, with arbitrary integer differences among the lower parameters. We express the product coefficients in each residue class as these rational functions times the coefficients of Henrici's known triple product. We obtain finite sums of ordinary ${}_8F_7$ functions whose lower parameters have positive real part, with each summand multiplied by a nonnegative integer power of the variable, and give an explicit formula when the first of three equal lower parameters is increased by one, together with bounds for the polynomial degrees and number of summands. We interpret the expressions for the three residue classes by taking limits of the entire summands when upper and lower parameters approach singular values together. We prove that the Taylor series has radius of convergence exactly $1/4$ whenever all original lower and upper parameters avoid nonpositive integers, without assuming integer differences among the lower parameters.

1. Introduction

For $u \in \mathbb{C}$ and $m \in \mathbb{Z}_{\geq 0}$, we put $(u)_0 = 1$ and $(u)_m = u(u+1)\cdots(u+m-1)$. We fix a positive integer k. For $\boldsymbol{m} = (m_1, \ldots, m_k) \in \mathbb{Z}_{\geq 0}^k$ and $\boldsymbol{x} = (x_1, \ldots, x_k) \in \mathbb{C}^k$, we put $|\boldsymbol{m}| = \sum_i m_i$, $\boldsymbol{m}! = \prod_i m_i!$, and $\boldsymbol{x}^{\boldsymbol{m}} = \prod_i x_i^{m_i}$. Lauricella's function F_C is defined by

$$F_C^{(k)}(a, b; c_1, \ldots, c_k; \boldsymbol{x}) = \sum_{\boldsymbol{m} \geq 0} \frac{(a)_{|\boldsymbol{m}|}(b)_{|\boldsymbol{m}|}}{\prod_i (c_i)_{m_i} \boldsymbol{m}!} \boldsymbol{x}^{\boldsymbol{m}},$$

where $a, b, c_1, \ldots, c_k$ are complex parameters and $c_i \notin \mathbb{Z}_{\leq 0}$. The series converges absolutely on $\sum_i \sqrt{|x_i|} < 1$. We also use the generalized hypergeometric series

$$(1.1) \quad {}_pF_q\left(\begin{matrix} u_1, \ldots, u_p \\ v_1, \ldots, v_q \end{matrix}; z\right) = \sum_{m \geq 0} \frac{\prod_{j=1}^p (u_j)_m}{\prod_{j=1}^q (v_j)_m} \frac{z^m}{m!}, \qquad {}_0F_1(; c; z) = \sum_{m \geq 0} \frac{z^m}{(c)_m m!}.$$

Here p, q are nonnegative integers, z is a complex variable, $u_1, \ldots, u_p, v_1, \ldots, v_q, c$ are complex parameters, and the lower parameters avoid $\mathbb{Z}_{\leq 0}$. We use ${}_pF_q$ with $p = q + 1$ as its ordinary series for $|z| < 1$ and then continue the germ along the specified path.

Henrici's triple product is expressed by a ${}_2F_7$ function in [Hen87], and related triple-series identities are studied in [KS90]. Reductions obtained by applying the beta integral to Henrici's product are given in [CWR13, Eq. (2.1) and its proof].

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This construction multiplies each coefficient by a factor depending on the total degree and keeps the three individual lower parameters equal. Here we use contiguous relations and the three ${}_2F_7$ expressions in (2.3) and (3.1)–(3.2) to express the coefficients after independent integer shifts as rational functions of the index times the coefficients in Henrici's product formula; see Theorem 4.1. We put $\omega = \exp(2\pi\sqrt{-1}/3)$ and define h_n by

$$\prod_{j=0}^{2} {}_0F_1(;c;\omega^j z) = \sum_{n\geq 0} h_n z^{3n}.$$

The explicit coefficients h_n follow from Henrici's product formula (2.3). Multiplication by $(a)_{3n}(b)_{3n}$ and triplication give the restriction with lower parameters c, c, c in Fact 2.2. Integer parameter shifts of the three-variable F_C system are reduced to eight differential components in [BK16, §3.1, (28), author manuscript arXiv:1602.00917v2]. That differential reduction of the three-variable system does not by itself express the coefficients on the one-variable restriction as rational multiples of the coefficients in Henrici's product formula. However, independent integer shifts can also produce the two other residue classes of the product coefficients. In this paper, we construct rational functions $R_r(n)$ for which the product coefficients equal $h_n R_r(n)$. We fix integers m_0, m_1, m_2 and consider the restriction

$$F_C^{(3)}(a, b; c+m_0, c+m_1, c+m_2; \kappa, \omega\kappa, \omega^2\kappa),$$

where κ is a complex variable and the original lower parameters satisfy $c+m_j \notin \mathbb{Z}_{\leq 0}$ for $j = 0, 1, 2$. For $m_0, m_1, m_2 \in \mathbb{Z}$, Theorem 4.1 gives unique rational functions $R_r \in \mathbb{Q}(\omega, c)(n)$ such that

$$[z^{3n+r}] \prod_{j=0}^{2} {}_0F_1(;c+m_j;\omega^j z) = h_n R_r(n) \qquad (0 \leq r \leq 2).$$

Here $[z^N]$ denotes the coefficient of z^N, and the equality first holds at parameters for which the displayed factors are finite. At other values of c satisfying $c + m_j \notin \mathbb{Z}_{\leq 0}$ for $j = 0, 1, 2$, we take the limit of the entire expression as c varies.

We choose a common lower parameter from which the three original lower parameters are obtained by nonnegative integer decreases, and apply the corresponding contiguous relations. Proposition 3.1 gives a finite sum of ordinary ${}_8F_7$ functions whose lower parameters have positive real part, with each summand multiplied by a nonnegative integer power of κ. Suppose that $c + m_j \notin \mathbb{Z}_{\leq 0}$ for $j = 0, 1, 2$ and $a, b \notin \mathbb{Z}_{\leq 0}$, so the upper parameters are nonterminating. Then Corollary 4.5 gives a first-order recurrence with rational functions of the index as coefficients, valid at all sufficiently large indices in each residue class whose coefficients are not eventually zero. These recurrences generate the first N Taylor coefficients in $O(N)$ arithmetic operations after fixed preprocessing, without factoring the numerator or denominator polynomials of the rational functions. Proposition 3.2 bounds the degrees of the polynomials in the recurrence that lowers the lower parameter and the number of summands. Proposition 3.3 gives all three residue classes explicitly for the lower parameters $c + 1, c, c$. We multiply coefficients by Pochhammer factors involving the upper parameters without dividing by these factors, so the formulas include terminating upper parameters.

The rational functions R_r determine the coefficient ratios of the restrictions. By factoring the numerator and denominator of R_r and forming pairs of upper and

lower parameters from their roots, we obtain the expression in Corollary 4.2. The number of summands records the three residue classes; the numbers of upper and lower parameters depend on the shifts. When $c + m_j \notin \mathbb{Z}_{\leq 0}$ for $j = 0, 1, 2$, we specify limits of the entire target expressions as c varies to resolve their apparent singularities. The finite ordinary ${}_8F_7$ sum gives an explicit continuation to $|\kappa| < 1/4$. The existence of this continuation also follows from the singular locus of the Lauricella differential system without assuming integer differences among the lower parameters; see Remark 4.6 and [HT14]. Proposition 4.7 shows that the Taylor series has radius of convergence exactly $1/4$ for arbitrary $c_0, c_1, c_2 \notin \mathbb{Z}_{\leq 0}$ and $a, b \notin \mathbb{Z}_{\leq 0}$, without assuming integer differences among the lower parameters.

The construction is related to the method in *Transformations of Lauricella's F_D and Correspondences between Hyperelliptic Curves* [Nak26]. In that paper, comparison of coefficients in a polynomial product gives an algebraic curve, whose rational coordinates lead to Lauricella representations through Mellin integrals. Here we use the product in (2.3) and the products obtained by applying contiguous relations to its factors to express the coefficients as rational functions times the coefficients in Henrici's product formula. After these shifts, the products need not be invariant under $z \mapsto \omega z$, so their coefficients can occur in all three residue classes. The rationality of these functions is a statement about coefficient sequences; it differs from the rationality of a function field in the F_D construction.

2. One-variable restrictions from a known triple product

In this section, we derive finite reductions of the restriction to $(\kappa, \omega\kappa, \omega^2\kappa)$ from Henrici's triple product [Hen87]. Related triple-series identities are studied in [KS90]. The reductions use contiguous relations and multiplication of the product coefficients. We use $1 + \omega + \omega^2 = 0$. Throughout this paper, a, b, c are complex parameters, and the original lower parameters avoid $\mathbb{Z}_{\leq 0}$. Apparent lower-parameter singularities in a displayed reduction are interpreted by varying all parameters depending on c together. For every nonnegative integer r, we define the parameter list

$$\Delta_r(a) = \left(\frac{a+r}{3}, \frac{a+r+1}{3}, \frac{a+r+2}{3} \right). \tag{2.1}$$

The following lemma gives the effect of multiplying the coefficient of degree n by $(a)_n(b)_n$.

Lemma 2.1. *Let $\phi(z) = \sum_{n\geq 0} u_n z^n$ be a formal power series satisfying*

$$A(\theta)\phi = \sum_{j=1}^{r} z^j B_j(\theta)\phi, \qquad \theta = z\frac{d}{dz},$$

where r is a nonnegative integer and $A, B_1, \ldots, B_r$ are polynomials. Then $Y(z) = \sum_{n\geq 0}(a)_n(b)_n u_n z^n$ satisfies

$$A(\theta)Y = \sum_{j=1}^{r} z^j B_j(\theta)(\theta + a)_j(\theta + b)_j Y. \tag{2.2}$$

The equation holds for arbitrary a, b, including terminating parameters.

PROOF. We compare the coefficient of z^n in the equation for ϕ. For $1 \le j \le \min(r,n)$, we use

$$(a)_n = (a)_{n-j}(a+n-j)_j, \qquad (b)_n = (b)_{n-j}(b+n-j)_j.$$

Terms with $j > n$ do not contribute to the coefficient of z^n. We multiply the coefficient equation by $(a)_n(b)_n$ and use these identities to obtain (2.2). No parameter division is required. □

2.1. **The known triple product and multiplication of its coefficients**

We introduce eight products indexed by subsets of $\{0,1,2\}$:

$$E_S(z) = \prod_{j\in S}\bigl(\theta\,{}_0F_1(;c;\omega^j z)\bigr)\prod_{j\notin S}{}_0F_1(;c;\omega^j z).$$

We put $g = c-1$, $\mathcal{H} = E_\varnothing$, $\mathcal{D} = E_{\{0,1,2\}}$, and

$$\mathcal{U}_r = \sum_{j=0}^{2}\omega^{rj}E_{\{j\}}, \qquad \mathcal{V}_r = \sum_{j=0}^{2}\omega^{rj}E_{\{0,1,2\}\setminus\{j\}} \quad (r=0,1,2).$$

These symbols occur in the differential system for the eight products E_S. The product identity in the following fact is Henrici's triple product theorem [Hen87]. The F_C identity follows by multiplying the coefficient of degree n by $(a)_n(b)_n$.

Fact 2.2. *For $c \notin \mathbb{Z}_{\le 0}$, we have*

$$\prod_{j=0}^{2}{}_0F_1(;c;\omega^j z)$$

$$= {}_2F_7\left(\begin{matrix}(2c-1)/4,(2c+1)/4\\ c,c/3,(c+1)/3,(c+2)/3,(2c-1)/3,2c/3,(2c+1)/3\end{matrix};\frac{64z^3}{729}\right) \tag{2.3}$$

for all $z \in \mathbb{C}$. As an identity of germs at the origin, we have

$$F_C^{(3)}(a,b;c,c,c;\kappa,\omega\kappa,\omega^2\kappa)$$

$$= {}_8F_7\left(\begin{matrix}\Delta_0(a),\Delta_0(b),(2c-1)/4,(2c+1)/4\\ c,c/3,(c+1)/3,(c+2)/3,(2c-1)/3,2c/3,(2c+1)/3\end{matrix};64\kappa^3\right). \tag{2.4}$$

The target series continues the restricted F_C germ from $|\kappa| < 1/9$ to $|\kappa| < 1/4$. At an apparent lower-parameter singularity, the entire right-hand sides have limits as c varies, with all occurrences of c varying together.

PROOF. We use the differential equation $\theta(\theta+g){}_0F_1(;c;\omega^j z) = \omega^j z\,{}_0F_1(;c;\omega^j z)$. The product rule gives

$$\theta E_S = \sum_{j\notin S}E_{S\cup\{j\}} + z\sum_{j\in S}\omega^j E_{S\setminus\{j\}} - |S|gE_S.$$

Taking the three Fourier sums yields the eight equations

$$\begin{aligned}
&\theta\mathcal{H} = \mathcal{U}_0, && (\theta+3g)\mathcal{D} = z\mathcal{V}_1,\\
&(\theta+g)\mathcal{U}_0 = 2\mathcal{V}_0, && (\theta+2g)\mathcal{V}_0 = 3\mathcal{D} - z\mathcal{U}_1,\\
&(\theta+g)\mathcal{U}_1 = -\mathcal{V}_1, && (\theta+2g)\mathcal{V}_1 = 2z\mathcal{U}_2,\\
&(\theta+g)\mathcal{U}_2 = 3z\mathcal{H} - \mathcal{V}_2, && (\theta+2g)\mathcal{V}_2 = -z\mathcal{U}_0.
\end{aligned} \tag{2.5}$$

We put

$$P_0(T) = T(T+g)(T+2g)(T+3g), \qquad T_0(T) = (T+g-1)(T+g)(T+2g-1)(T+2g).$$

Eliminating $\mathcal{U}_0, \mathcal{V}_0, \mathcal{D}$ gives

$$P_0(\theta)\mathcal{H} = -2z(4\theta + 1 + 6g)\mathcal{U}_1.$$

In fact, we substitute $\mathcal{V}_0 = (\theta + g)\theta\mathcal{H}/2$, $3\mathcal{D} = (\theta + 2g)\mathcal{V}_0 + z\mathcal{U}_1$, and $(\theta + 3g)\mathcal{D} = -z(\theta + g)\mathcal{U}_1$. Eliminating $\mathcal{V}_1, \mathcal{U}_2, \mathcal{V}_2$ gives

$$T_0(\theta)\mathcal{U}_1 = -2z^2(4\theta + 3 + 6g)\mathcal{H}.$$

For this elimination, we use

$$2z\mathcal{U}_2 = -(\theta + 2g)(\theta + g)\mathcal{U}_1, \qquad (\theta + 2g)(\theta + g)\mathcal{U}_2 = z(4\theta + 3 + 6g)\mathcal{H}.$$

We apply $T_0(\theta - 1)$ to the first eliminated equation and use $\theta z = z(\theta + 1)$ to obtain

$$P_0(\theta)T_0(\theta - 1)\mathcal{H} = 64z^3 \left(\theta + \frac{3(2c - 1)}{4}\right)\left(\theta + \frac{3(2c + 1)}{4}\right)\mathcal{H}.$$

Thus

$$\begin{aligned}\Bigg[\theta(\theta + 3c - 3)\prod_{j=1}^{3}(\theta + c - j)\prod_{j=2}^{4}(\theta + 2c - j) \\ - 64z^3 \left(\theta + \frac{3(2c - 1)}{4}\right)\left(\theta + \frac{3(2c + 1)}{4}\right)\Bigg]\mathcal{H} = 0.\end{aligned} \tag{2.6}$$

All eliminations are polynomial identities; no division by a variable parameter occurs.

Since $\mathcal{H}(\omega z) = \mathcal{H}(z)$ and $\mathcal{H}(0) = 1$, we write $\mathcal{H} = \sum_{n\geq 0} h_n z^{3n}$. At values of c for which the coefficient ratio is defined, comparison in (2.6) gives

$$\frac{h_n}{h_{n-1}} = \frac{64}{729}\frac{(n - 1 + (2c - 1)/4)(n - 1 + (2c + 1)/4)}{n(n - 1 + c)\prod_{j=0}^{2}(n - 1 + (c + j)/3)\prod_{j=0}^{2}(n - 1 + (2c - 1 + j)/3)}. \tag{2.7}$$

This is the coefficient ratio in (2.3). Both sides are entire in z, so coefficient comparison proves that identity. Their coefficients are meromorphic in c, and the product on the source side is holomorphic for $c \notin \mathbb{Z}_{\leq 0}$. Hence the apparent singularities on the target side are removable when the entire expression is regarded as a function of c.

For $|\kappa| < 1/9$, we group the absolutely convergent F_C series by total degree. Its restriction is

$$\sum_{n\geq 0}(a)_{3n}(b)_{3n}h_n\kappa^{3n}.$$

We use

$$(a)_{3n} = 3^{3n}(a/3)_n((a + 1)/3)_n((a + 2)/3)_n$$

and the same identity for b. The resulting factor $3^{6n} = 729^n$ changes the target argument from $64z^3/729$ to $64\kappa^3$. This proves (2.4) without a real-part condition on a, b. The target ${}_8F_7$ series converges for $|64\kappa^3| < 1$. With $\theta = \kappa\, d/d\kappa$, Lemma 2.1 also gives the continued differential equation by multiplying the shifted term in (2.6) by $\prod_{j=0}^{2}(\theta + a + j)(\theta + b + j)$. The coefficient of θ^8 in this equation is $1 - 64\kappa^3$. After leaving the origin, we continue along paths that avoid $\kappa = 0$ and the three zeros of $1 - 64\kappa^3$. This continuation preserves holomorphic dependence on every $c \notin \mathbb{Z}_{\leq 0}$. Thus the limits of the entire expressions as c varies agree with the continued F_C germ on the larger disk and on any specified common continuation path. □

Remark 2.3. At $c = 1/2$, we obtain $3/4$ as the limit of the ratio of the Pochhammer factor for the upper parameter $(2c-1)/4$ to that for the lower parameter $(2c-1)/3$ in every coefficient of positive degree. Consequently,

$$F_C^{(3)}(a,b;1/2,1/2,1/2;\kappa,\omega\kappa,\omega^2\kappa) = \frac{1}{4} + \frac{3}{4}\,{}_6F_5\left(\begin{matrix}\Delta_0(a),\Delta_0(b)\\ 1/6,1/3,1/2,2/3,5/6\end{matrix};64\kappa^3\right).$$

An isolated cancellation of equal zero parameters would omit the constant $1/4$ and does not give this limit.

3. Integer shifts of the lower parameters

In this section, we compute $\mathcal{U}_1$,$\mathcal{U}_2$ and apply contiguous relations to shift each lower parameter by an integer. The same differential system gives explicit expressions for $\mathcal{U}_1$ and $\mathcal{U}_2$. We have

$$\mathcal{U}_1 = -\frac{3z^2}{c^2(c+1)}\,{}_2F_7\left(\begin{matrix}(2c+1)/4,(2c+3)/4\\ c,(c+2)/3,(c+3)/3,(c+4)/3,\\ (2c+1)/3,(2c+2)/3,(2c+3)/3\end{matrix};\frac{64z^3}{729}\right), \tag{3.1}$$

$$\mathcal{U}_2 = \frac{3z}{c}\,{}_2F_7\left(\begin{matrix}(2c+1)/4,(2c+3)/4\\ c,(c+1)/3,(c+2)/3,(c+3)/3,\\ 2c/3,(2c+1)/3,(2c+2)/3\end{matrix};\frac{64z^3}{729}\right). \tag{3.2}$$

To verify these formulas, we write their coefficients of z^{3n+2} and z^{3n+1} as b_n, d_n, respectively. The eliminated equations in the proof of Fact 2.2 give

$$\begin{aligned}\frac{b_n}{h_n} &= \frac{-6(4n+2c-1)}{(3n+c)(3n+c+1)(3n+2c-1)(3n+2c)},\\ \frac{d_n}{h_n} &= \frac{3(4n+2c-1)}{(3n+c)(3n+2c-1)}.\end{aligned} \tag{3.3}$$

Their initial values are $-3/[c^2(c+1)]$ and $3/c$. Substitution of (2.7) gives the coefficient ratios and initial values in (3.1)–(3.2). The meromorphic identity in c then gives the limits of the entire expressions as c varies.

We fix the order of each target function in the following proposition by choosing a common lower parameter from which the three original lower parameters are obtained by nonnegative integer decreases.

Proposition 3.1. *For fixed $m_0, m_1, m_2 \in \mathbb{Z}$, the germ at the origin of*

$$F_C^{(3)}(a,b;c+m_0,c+m_1,c+m_2;\kappa,\omega\kappa,\omega^2\kappa)$$

is a finite sum of nonnegative integer powers of κ times ordinary ${}_8F_7$ functions. We obtain the parameters and coefficients from the recursion (3.5) *and the three ${}_2F_7$ expressions in* (2.3) *and* (3.1)–(3.2)*. At each step of the recursion, we decrease the lower parameter by one. We then multiply coefficients as in* (3.6)*. The formula holds for independent a, b, c satisfying $c + m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$. The finite sum continues the germ at the origin from $|\kappa| < 1/9$ to $|\kappa| < 1/4$. The common lower parameter can be chosen so that every lower parameter of the resulting ${}_8F_7$ functions has positive real part.*

Proof. We choose a nonnegative integer h and put

$$M_* = \max(m_0,m_1,m_2), \qquad C_* = c + M_* + h, \qquad d_j = M_* + h - m_j \ge 0, \tag{3.4}$$

with $\operatorname{Re} C_* > 1/2$. Such a choice is possible at every value of c for which $c + m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$, and throughout a neighborhood of that value. For $\theta = x\, d/dx$, we define polynomials $\widehat{A}_d, \widehat{B}_d$ by $\widehat{A}_0 = 1, \widehat{B}_0 = 0$ and

$$\begin{aligned} \widehat{A}_{d+1} &= \widehat{A}_d + \frac{\theta \widehat{A}_d + x \widehat{B}_d}{C_* - d - 1}, \\ \widehat{B}_{d+1} &= \widehat{B}_d + \frac{\widehat{A}_d + \theta \widehat{B}_d + (1 - C_*) \widehat{B}_d}{C_* - d - 1}. \end{aligned} \tag{3.5}$$

The backward contiguous relation and the equation $\theta^2 {}_0F_1(;C_*;x) = x\, {}_0F_1(;C_*;x) + (1 - C_*)\theta\, {}_0F_1(;C_*;x)$ give

$${}_0F_1(;C_* - d;x) = \widehat{A}_d(x)\, {}_0F_1(;C_*;x) + \widehat{B}_d(x)\, \theta\, {}_0F_1(;C_*;x).$$

No negative powers of x occur in this recursion. For $0 \le d < d_j$, a zero denominator would imply $C_* - d_j = -(d_j - d - 1) \in \mathbb{Z}_{\le 0}$, contrary to $c + m_j = C_* - d_j \notin \mathbb{Z}_{\le 0}$.

We apply this representation to the three factors with $x = \omega^j z$. The coefficient multiplying each product E_S, with $c = C_*$, is a polynomial in z. Fourier inversion and (2.5) express every E_S through powers of θ applied to (2.3) and (3.1)–(3.2), using nonnegative powers of z. Writing these polynomials in θ in terms of falling factorials gives a finite sum of terms

$$D z^s\, {}_2F_7\left(\begin{matrix} u_1, u_2 \\ v_1, \dots, v_7 \end{matrix}; \frac{64 z^3}{729}\right), \qquad s \ge 0.$$

Here D is a scalar coefficient obtained from the recursion, and the parameter lists are those of the three displayed ${}_2F_7$ functions at $c = C_*$, with the same nonnegative integer added to every upper and lower parameter by differentiation. All v_j have positive real part. For each term, multiplication of the coefficient of z^n by $(a)_n (b)_n$ gives

$$\begin{aligned} &\sum_{n \ge 0} \frac{(u_1)_n (u_2)_n (64/729)^n}{n! \prod_{j=1}^{7} (v_j)_n} (a)_{3n+s} (b)_{3n+s} \kappa^{3n+s} \\ &\qquad = (a)_s (b)_s \kappa^s\, {}_8F_7\left(\begin{matrix} u_1, u_2, \Delta_s(a), \Delta_s(b) \\ v_1, \dots, v_7 \end{matrix}; 64\kappa^3\right). \end{aligned} \tag{3.6}$$

This identity follows from triplication and requires no division by $(a)_s$ or $(b)_s$, so it includes terminating upper parameters. We justify the coefficient comparison by the absolute convergence of the source series on $|\kappa| < 1/9$. The finite sum of ordinary ${}_8F_7$ series is holomorphic on $|\kappa| < 1/4$. This proves the proposition, including parameters that would give apparent singularities with a smaller common lower parameter.

To express the coefficients as rational functions of the index times h_n, we also record a construction with the fixed original lower parameter c. We first take values of c at which every denominator in the following finite recursions is nonzero. For an integer m, we use $\theta = x\, d/dx$ and construct Laurent polynomials $A_m(x), B_m(x)$, with coefficients rational in c, satisfying

$$\begin{aligned} {}_0F_1(;c + m;x) = A_m(x)\, {}_0F_1(;c;x) \\ + B_m(x)\, \theta\, {}_0F_1(;c;x). \end{aligned}$$

We start with $A_0 = 1, B_0 = 0$. The differential equation for ${}_0F_1(;c;x)$ and the identities

$$\begin{aligned} {}_0F_1(;c+m+1;x) &= \frac{c+m}{x}\theta\,{}_0F_1(;c+m;x),\\ {}_0F_1(;c+m-1;x) &= {}_0F_1(;c+m;x)\\ &\quad + \frac{\theta\,{}_0F_1(;c+m;x)}{c+m-1} \end{aligned}$$

give the finite forward and backward recursions

$$\begin{aligned} A_{m+1} &= \frac{c+m}{x}(\theta A_m + xB_m),\\ B_{m+1} &= \frac{c+m}{x}(A_m + \theta B_m + (1-c)B_m), \end{aligned}$$

and

$$\begin{aligned} A_{m-1} &= A_m + \frac{\theta A_m + xB_m}{c+m-1},\\ B_{m-1} &= B_m + \frac{A_m + \theta B_m + (1-c)B_m}{c+m-1}. \end{aligned}$$

Thus any integer offset requires only finitely many operations.

We substitute $x = \omega^j z$ into these expressions, use $\theta = z\,d/dz$, and multiply the three factors. The result is a finite linear combination of the eight products E_S, with Laurent polynomials as coefficients. Fourier inversion gives

$$E_{\{j\}} = \frac{1}{3}(\mathcal{U}_0 + \omega^{-j}\mathcal{U}_1 + \omega^{-2j}\mathcal{U}_2),$$

$$E_{\{0,1,2\}\setminus\{j\}} = \frac{1}{3}(\mathcal{V}_0 + \omega^{-j}\mathcal{V}_1 + \omega^{-2j}\mathcal{V}_2).$$

The equations (2.5) also give

$$\mathcal{U}_0 = \theta\mathcal{H}, \qquad \mathcal{V}_0 = \tfrac{1}{2}(\theta+g)\theta\mathcal{H}, \qquad \mathcal{V}_1 = -(\theta+g)\mathcal{U}_1,$$

$$\mathcal{V}_2 = 3z\mathcal{H} - (\theta+g)\mathcal{U}_2, \qquad \mathcal{D} = \tfrac{1}{6}(\theta+2g)(\theta+g)\theta\mathcal{H} + \tfrac{z}{3}\mathcal{U}_1.$$

Hence every term is a monomial times a polynomial in θ of degree at most three acting on one of the three explicit ${}_2F_7$ expressions.

We put $w = 64z^3/729$ and $\Theta = w\,d/dw$. On a term $z^s\phi(w)$, the operator θ becomes $s + 3\Theta$. We expand powers of Θ in the falling factorials $\Theta(\Theta-1)\cdots(\Theta-r+1) = w^r(d/dw)^r$. Termwise differentiation of (1.1) gives

$$\left(\frac{d}{dw}\right)^r {}_2F_7\left(\begin{matrix}u_1,u_2\\ v_1,\dots,v_7\end{matrix};w\right) = \frac{(u_1)_r(u_2)_r}{\prod_{j=1}^7 (v_j)_r}\,{}_2F_7\left(\begin{matrix}u_1+r,u_2+r\\ v_1+r,\dots,v_7+r\end{matrix};w\right).$$

We therefore obtain a finite sum of terms

$$Az^t\,{}_2F_7\left(\begin{matrix}u_1,u_2\\ v_1,\dots,v_7\end{matrix};\frac{64z^3}{729}\right), \qquad t\in\mathbb{Z},$$

with all lists determined by the recursions.

If $t<0$, we put $N = \lceil -t/3\rceil$; otherwise we put $N=0$. We remove the first N Taylor terms in that expression and put

$$U_N = \frac{(64/729)^N (u_1)_N (u_2)_N}{N!\prod_{j=1}^7 (v_j)_N}, \qquad r = t+3N \geq 0.$$

The remaining term is

$$AU_N z^r {}_3F_8\left(\begin{matrix}u_1+N, u_2+N, 1\\ v_1+N, \dots, v_7+N, N+1\end{matrix}; \frac{64z^3}{729}\right).$$

For $N=0$, we cancel the extra upper and lower parameters 1, leaving ${}_2F_7$. Every removed term has a negative integer power of z. The original product of three ${}_0F_1$ functions is holomorphic at zero, so the sum of the removed negative powers is zero coefficient by coefficient. Thus the series remaining after the negative powers are removed reproduce the original product.

We multiply each coefficient of z^n by $(a)_n(b)_n$. For one of these remaining series

$$Az^r {}_pF_q\left(\begin{matrix}u_1, \dots, u_p\\ v_1, \dots, v_q\end{matrix}; 64z^3/729\right),$$

with $(p,q)=(2,7)$ or $(3,8)$, the result is

$$A(a)_r(b)_r\kappa^r {}_{p+6}F_q\left(\begin{matrix}u_1, \dots, u_p, \Delta_r(a), \Delta_r(b)\\ v_1, \dots, v_q\end{matrix}; 64\kappa^3\right). \tag{3.7}$$

Here A includes the factor U_N when the first N Taylor terms have been removed. The identity follows from the triplication formula for $(a)_{3n+r}$ and $(b)_{3n+r}$. It preserves terminating upper parameters, since it uses no division by them. The resulting orders are ${}_8F_7$ and ${}_9F_8$, and every step produces finitely many terms. Coefficient comparison on $|\kappa|<1/9$ gives the restricted F_C series. The resulting generalized hypergeometric series converge on $|\kappa|<1/4$. Finally, the identity is meromorphic in c. For every c satisfying $c+m_j \notin \mathbb{Z}_{\le 0}$ for $j=0,1,2$, the limit of the entire expression as c varies agrees with the original germ and its continuation, as in Fact 2.2. □

The recursion (3.5), which decreases the lower parameter by one at each step, also gives bounds for the degrees and the number of summands.

Proposition 3.2. *For the polynomials in* (3.5)*, we have*

$$\deg \widehat{A}_d \le \left\lfloor \frac{d}{2} \right\rfloor, \qquad \deg \widehat{B}_d \le \left\lfloor \frac{d-1}{2} \right\rfloor, \tag{3.8}$$

where $d \ge 0$ *and* $\deg 0 = -\infty$*. For* d_j, C_* *as in* (3.4)*, we put* $D_{\mathrm{tot}} = d_0+d_1+d_2$*. The construction in Proposition 3.1 uses at most*

$$35\left(\left\lfloor \frac{D_{\mathrm{tot}}}{2} \right\rfloor + 1\right) \tag{3.9}$$

ordinary ${}_8F_7$ *summands before we combine equal parameter lists. The bound depends on the chosen integer* h *through* D_{tot}*.*

Proof. We use induction on d. The initial polynomials $\widehat{A}_0=1$ and $\widehat{B}_0=0$ satisfy (3.8). Since applying θ does not increase polynomial degree, (3.5) gives

$$\deg \widehat{A}_{d+1} \le \max\left(\left\lfloor \frac{d}{2} \right\rfloor, 1+\left\lfloor \frac{d-1}{2} \right\rfloor\right) = \left\lfloor \frac{d+1}{2} \right\rfloor,$$
$$\deg \widehat{B}_{d+1} \le \max\left(\left\lfloor \frac{d}{2} \right\rfloor, \left\lfloor \frac{d-1}{2} \right\rfloor\right) = \left\lfloor \frac{d}{2} \right\rfloor.$$

For each $S \subset \{0,1,2\}$, the polynomial multiplying E_S is

$$\prod_{j \notin S} \widehat{A}_{d_j}(\omega^j z) \prod_{j \in S} \widehat{B}_{d_j}(\omega^j z).$$

By (3.8), its degree is at most $\lfloor D_{\mathrm{tot}}/2 \rfloor$. Fourier inversion and (2.5) express the eight products through $\mathcal{H}, \mathcal{U}_1, \mathcal{U}_2$. We put $w = 64z^3/729$ and $\Theta = w\, d/dw$, as in the proof of Proposition 3.1. For the expression for $\mathcal{H}$ in (2.3), the operator θ becomes 3Θ. Thus a polynomial in θ with zero constant term has no derivative of order zero when expanded in the falling factorials displayed above. Thus the expression for $\mathcal{H}$ contains one term involving an ordinary derivative, and the expression for each $E_{\{j\}}$, $j = 0,1,2$, contains three such terms. For each product indexed by a two-element subset, $\mathcal{V}_0 = (\theta+g)\theta\mathcal{H}/2$ requires two terms, $\mathcal{V}_1 = -(\theta+g)\mathcal{U}_1$ requires two, and $\mathcal{V}_2 = 3z\mathcal{H} - (\theta+g)\mathcal{U}_2$ requires three. The triple product $\mathcal{D} = (\theta+2g)(\theta+g)\theta\mathcal{H}/6 + z\mathcal{U}_1/3$ requires four. The total is $1 + 3\cdot 3 + 3\cdot 7 + 4 = 35$. Each polynomial multiplying a product E_S has at most $\lfloor D_{\mathrm{tot}}/2 \rfloor + 1$ monomials, which proves (3.9). Multiplying coefficients as in (3.6) preserves the number of summands. □

We give an explicit reduction for lower parameters $c+1, c, c$.

Proposition 3.3. *Suppose that $c, c+1 \notin \mathbb{Z}_{\le 0}$. We use the upper parameter lists*

$$\boldsymbol{A}_0 = \boldsymbol{A}_1 = \left(\frac{2c+1}{4}, \frac{2c+3}{4}\right),$$
$$\boldsymbol{A}_2 = \left(\frac{2c+3}{4}, \frac{2c+5}{4}\right),$$

and the lower parameter lists

$$\boldsymbol{B}_0 = \left(c, \frac{c+1}{3}, \frac{c+2}{3}, \frac{c+3}{3}, \frac{2c}{3}, \frac{2c+1}{3}, \frac{2c+2}{3}\right),$$
$$\boldsymbol{B}_1 = \left(c, \frac{c+2}{3}, \frac{c+3}{3}, \frac{c+4}{3}, \frac{2c+1}{3}, \frac{2c+2}{3}, \frac{2c+3}{3}\right),$$
$$\boldsymbol{B}_2 = \left(c+1, \frac{c+3}{3}, \frac{c+4}{3}, \frac{c+5}{3}, \frac{2c+2}{3}, \frac{2c+3}{3}, \frac{2c+4}{3}\right).$$

For independent a, b, we have

$$\begin{aligned} &F_C^{(3)}(a,b;c+1,c,c;\kappa,\omega\kappa,\omega^2\kappa) \\ &\quad = {}_8F_7\left(\begin{matrix}\Delta_0(a), \Delta_0(b), \boldsymbol{A}_0 \\ \boldsymbol{B}_0\end{matrix}; 64\kappa^3\right) \\ &\qquad - \frac{ab\kappa}{c(c+1)}\,{}_8F_7\left(\begin{matrix}\Delta_1(a), \Delta_1(b), \boldsymbol{A}_1 \\ \boldsymbol{B}_1\end{matrix}; 64\kappa^3\right) \\ &\qquad + \frac{2(a)_2(b)_2\kappa^2}{c^2(c+1)(c+2)}\,{}_8F_7\left(\begin{matrix}\Delta_2(a), \Delta_2(b), \boldsymbol{A}_2 \\ \boldsymbol{B}_2\end{matrix}; 64\kappa^3\right). \end{aligned} \tag{3.10}$$

Here Δ_r is the list of three parameters defined in (2.1). *The equality first holds as an identity of germs at the origin. The right-hand side continues the germ to $|\kappa| < 1/4$, with the limit of the entire expression as c varies at apparent lower-parameter singularities.*

PROOF. We use ${}_0F_1(;c+1;z) = c\theta\,{}_0F_1(;c;z)/z$ and Fourier inversion to obtain

$$\prod_{j=0}^{2} {}_0F_1(;c+\mathbf{1}_{j=0};\omega^j z) = \frac{c}{3z}\big(\theta\mathcal{H} + \mathcal{U}_1 + \mathcal{U}_2\big).$$

Here $\mathbf{1}_{j=0}$ equals one for $j=0$ and zero for $j=1,2$. The formulas (3.1)–(3.2) give the first two residue classes. Since $h_1 = 2/[c^3(c+1)(c+2)]$, differentiation of (2.3) gives the remaining class:

$$\frac{c}{3z}\theta\mathcal{H} = \frac{2z^2}{c^2(c+1)(c+2)}\,{}_2F_7\left(\begin{matrix} \frac{2c+3}{4}, \frac{2c+5}{4} \\ c+1, \frac{c+3}{3}, \frac{c+4}{3}, \frac{c+5}{3} \\ \frac{2c+2}{3}, \frac{2c+3}{3}, \frac{2c+4}{3} \end{matrix}; \frac{64z^3}{729}\right).$$

We apply (3.6) to the three terms and obtain (3.10). The coefficient identities include terminating upper parameters. The limits of the entire expressions as c varies follow as in Fact 2.2. □

4. Rational functions relating the coefficients

In this section, we construct rational functions $R_r(n)$ that compare the product coefficients with the coefficients in Henrici's product formula, and express the resulting coefficient ratios in terms of hypergeometric parameters. We put

$$\boldsymbol{A} = ((2c-1)/4, (2c+1)/4),$$
$$\boldsymbol{D} = (c, c/3, (c+1)/3, (c+2)/3, (2c-1)/3, 2c/3, (2c+1)/3).$$

The coefficients in (2.3) are

$$h_n = \frac{(64/729)^n \prod_{i=1}^{2}(A_i)_n}{n!\prod_{j=1}^{7}(D_j)_n}.$$

Scalar addition and subtraction act on every entry of a parameter list.

The recursions that shift the lower parameters by integers preserve rational multiples of the coefficients h_n.

Theorem 4.1. *For each fixed $m_0, m_1, m_2 \in \mathbb{Z}$, the recursions in Proposition 3.1 construct rational functions $R_r \in \mathbb{Q}(\omega, c)(n)$, $0 \le r \le 2$, such that*

$$\prod_{j=0}^{2} {}_0F_1(;c+m_j;\omega^j z) = \sum_{r=0}^{2} z^r \sum_{n\ge 0} h_n R_r(n) z^{3n}. \tag{4.1}$$

The equality first holds at values of c for which the displayed coefficients are finite and the coefficient ratios used in the construction are defined. The rational functions R_r are unique for the fixed sequence h_n. At other c satisfying $c + m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$, we take the meromorphic limit of the entire expression as c varies.

PROOF. The proof of Proposition 3.1 expresses the product as a finite sum of terms

$$Cz^e P(\theta)\left\{ z^s \sum_{k\ge 0} h_k V_s(k) z^{3k} \right\}, \qquad s \in \{0,1,2\}.$$

Here $V_0(n) = 1$, and $V_1(n) = d_n/h_n$ and $V_2(n) = b_n/h_n$ are given by (3.3). The index s records the exponent in z^s, rather than the index on $\mathcal{U}_r$ and $\mathcal{V}_r$. These

rational functions V_s differ from the functions $\mathcal{V}_s$. For $e+s = r+3\ell$, the contribution to $R_r(n)$ is $C(h_{n-\ell}/h_n)V_s(n-\ell)P(3(n-\ell)+s)$. For $\ell > 0$, we have

$$\frac{h_{n-\ell}}{h_n} = \left(\frac{64}{729}\right)^{-\ell} \frac{(n-\ell+1)_\ell \prod_{j=1}^{7}(D_j+n-\ell)_\ell}{\prod_{i=1}^{2}(A_i+n-\ell)_\ell}.$$

When the denominator factors are nonzero, this rational expression vanishes at $0 \le n < \ell$, recovering the coefficients that are absent because their indices are negative. For $\ell \le 0$, the coefficient recurrence gives a rational expression for the same ratio. Summing these finitely many contributions proves (4.1) for every $n \ge 0$. The construction from the fixed lower parameter c uses only rational operations in ω, c, n, so $R_r \in \mathbb{Q}(\omega, c)(n)$. At values of c for which all coefficients are finite and $h_n \ne 0$ for every $n \ge 0$, any two choices of R_r agree at every nonnegative integer. A nonzero rational function has only finitely many zeros, so the rational functions coincide. The original product is holomorphic whenever $c+m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$, which gives the limit of the entire expression as c varies. □

For each residue class, we define the coefficient

$$t_{r,n} = (a)_{3n+r}(b)_{3n+r}h_n R_r(n).$$

These are the Taylor coefficients of the restricted F_C function at degrees $3n+r$. At parameter values and indices for which all displayed factors are finite and $t_{r,n} \ne 0$, we have

$$\frac{t_{r,n+1}}{t_{r,n}} = 64\, \frac{\displaystyle\prod_{i=1}^{2}(n+A_i) \prod_{j=0}^{2}\left(n+\frac{a+r+j}{3}\right) \prod_{j=0}^{2}\left(n+\frac{b+r+j}{3}\right)}{(n+1)\displaystyle\prod_{j=1}^{7}(n+D_j)} \frac{R_r(n+1)}{R_r(n)}.$$

After we clear denominators, the recurrence remains valid at zero coefficients but does not necessarily determine the next coefficient there. For coefficient generation, we use the finite initial segment specified in Corollary 4.5. The roots of the numerator and denominator of R_r specify pairs of upper and lower parameters that encode this coefficient sequence. We obtain the following expression by allowing the numbers of upper and lower parameters to depend on the shifts.

Corollary 4.2. *Let R_r be the rational functions in Theorem 4.1. We first take values of c for which $c+m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$, all lower entries of $\boldsymbol{D}$ avoid $\mathbb{Z}_{\le 0}$, and h_n and $R_r(n)$ are finite with $h_n \ne 0$ for every $n \ge 0$. For $R_r \not\equiv 0$, we write $R_r = P_r/Q_r$ in reduced form and use the lists of roots of P_r and Q_r,*

$$\boldsymbol{\rho}_r = (\rho_{r,1}, \dots, \rho_{r,u_r}), \qquad \boldsymbol{\sigma}_r = (\sigma_{r,1}, \dots, \sigma_{r,v_r}),$$

with multiplicities. We put $N_r = \min\{n \ge 0 : R_r(n) \ne 0\}$ and $s_r = r + 3N_r$. Then

$$\begin{aligned} &F_C^{(3)}(a, b; c+m_0, c+m_1, c+m_2; \kappa, \omega\kappa, \omega^2\kappa) \\ &= \sum_{\substack{0 \le r \le 2 \\ R_r \not\equiv 0}} \lim_{\epsilon \to 0} \Bigg[h_{N_r} R_r(N_r+\epsilon)(a)_{s_r}(b)_{s_r}\kappa^{s_r} \end{aligned}$$

$$\times\, {}_{9+u_r+v_r}F_{8+u_r+v_r}\left(\begin{matrix} 1, \boldsymbol{A}+N_r, \Delta_{s_r}(a), \Delta_{s_r}(b), \\ N_r+\epsilon+1-\boldsymbol{\rho}_r,\ N_r+\epsilon-\boldsymbol{\sigma}_r \\ N_r+1, \boldsymbol{D}+N_r, \\ N_r+\epsilon-\boldsymbol{\rho}_r,\ N_r+\epsilon+1-\boldsymbol{\sigma}_r \end{matrix}; 64\kappa^3\right)\Bigg]. \tag{4.2}$$

The equality holds for independent a, b on $|\kappa| < 1/9$, and its right-hand side continues the germ at the origin to $|\kappa| < 1/4$. The limit in each summand uses the same ϵ in every pair of upper and lower parameters and includes the whole prefactor. It can be replaced by direct substitution if all lower parameters of the generalized hypergeometric function at $\epsilon = 0$ avoid $\mathbb{Z}_{\le 0}$. At other c satisfying $c + m_j \notin \mathbb{Z}_{\le 0}$ for $j = 0, 1, 2$, the formula is the meromorphic limit in c of the entire sum.

Proof. We factor the reduced numerator and denominator and obtain the exact identity

$$\frac{R_r(N_r+n+\epsilon)}{R_r(N_r+\epsilon)} = \prod_{i=1}^{u_r} \frac{(N_r+\epsilon+1-\rho_{r,i})_n}{(N_r+\epsilon-\rho_{r,i})_n} \prod_{j=1}^{v_r} \frac{(N_r+\epsilon-\sigma_{r,j})_n}{(N_r+\epsilon+1-\sigma_{r,j})_n}.$$

The upper parameter 1 cancels $n!$. The lower parameter $N_r + 1$ supplies the factorial in h_{N_r+n}/h_{N_r}. Thus the coefficient operation in (3.7) gives (4.2) coefficient by coefficient. After multiplication by the prefactor, the coefficient with index n tends to $h_{N_r+n} R_r(N_r+n)(a)_{3n+s_r}(b)_{3n+s_r}$. The rational functions $R_r(N_r+n+\epsilon)$ have polynomial growth in n, uniformly for sufficiently small ϵ, because their denominator roots stay away from these integers. When $a, b \notin \mathbb{Z}_{\le 0}$, the remaining coefficient ratios tend to 64; the resulting polynomial bounds also hold when an upper parameter is a nonpositive integer. Thus the series and their limits converge locally uniformly on $|\kappa| < 1/4$. This justifies the limits as $\epsilon \to 0$ and the continuation; meromorphic dependence on c gives the final assertion. □

Remark 4.3. For $N_r = 0$, the upper and lower parameters 1 can be canceled. The limit as $\epsilon \to 0$ can still be necessary when all original lower parameters avoid $\mathbb{Z}_{\le 0}$. For the offsets $(2, 0, 0)$, the rational function R_2 is

$$R_2(n) = \frac{4c(c+1)(2c+4n-1)(2c+4n+1)(5c+9n+6)}{\left[\begin{matrix} (c+n)(c+3n)(2c+3n)(c+3n+1)(c+3n+2) \\ \times(c+3n+3)(2c+3n-1)(2c+3n+1)(2c+3n+2) \end{matrix}\right]}. \tag{4.3}$$

We use the contiguous relation

$${}_0F_1(;c+2;z) = \frac{c(c+1)}{z}\,{}_0F_1(;c;z) - \frac{c^2(c+1)}{z^2}\theta\,{}_0F_1(;c;z),$$

and (3.3) to obtain

$$R_2(n) = c(c+1)\frac{h_{n+1}}{h_n}\left(1 - \frac{c}{3}\frac{d_{n+1}}{h_{n+1}}\right),$$

which gives (4.3) by (2.7). For $(m_0, m_1, m_2) = (2, 0, 0)$ and $c = -24/5$, the construction gives $N_2 = 0$, $R_2(2) = 0$, and $R_2(3) = -1009375/304668$. The numerator root 2 contributes the upper parameter $-1+\epsilon$ and lower parameter $-2+\epsilon$. Treating the upper parameter -1 as an isolated termination would discard the nonzero coefficient with index 3. The limit as $\epsilon \to 0$ in (4.2) preserves the nonzero coefficient with index 3.

The rational functions giving the ratios of Taylor coefficients determine the radius of convergence whenever $c+m_j \notin \mathbb{Z}_{\le 0}$ for $j=0,1,2$ and the upper parameters are nonterminating.

Corollary 4.4. *Suppose that $m_0,m_1,m_2 \in \mathbb{Z}$, $c+m_j \notin \mathbb{Z}_{\le 0}$ for $j=0,1,2$, and $a,b \notin \mathbb{Z}_{\le 0}$. Then the Taylor series at $\kappa=0$ of*

$$F_C^{(3)}(a,b;c+m_0,c+m_1,c+m_2;\kappa,\omega\kappa,\omega^2\kappa)$$

has radius of convergence exactly $1/4$.

Proof. We choose the common lower parameter $C_* = c+M_*+h$ in (3.4) with $\operatorname{Re} C_* > 1/2$. All coefficients $h_n(C_*)$ of (2.3) are finite and nonzero. The recursion (3.5) has no zero parameter denominator, since every intermediate lower parameter differs by a nonnegative integer from an original lower parameter that avoids $\mathbb{Z}_{\le 0}$. For fixed c,m_0,m_1,m_2, Fourier inversion and finite index shifts therefore give rational functions $S_r \in \mathbb{C}(n)$ such that

$$[z^{3n+r}]\prod_{j=0}^{2} {}_0F_1(;c+m_j;\omega^j z) = h_n(C_*)S_r(n) \qquad (n \ge N_0,\ 0 \le r \le 2) \tag{4.4}$$

for some nonnegative integer N_0. All ratios of shifted coefficients in Henrici's product formula are rational functions of n, and any possible integer zero of their denominators is avoided for sufficiently large n. This argument at sufficiently large indices does not specialize a rational function constructed from the fixed lower parameter c at a small exceptional index.

The three functions S_r cannot all be zero. To see this directly, we put $\gamma_j = c+m_j$ and use principal powers on the rays with arguments $0, 2\pi/3, -2\pi/3$. The relation to the modified Bessel function I_ν of the first kind and its asymptotic expansion are given in [DLMF, Eqs. 10.39.9 and 10.40.1]; see also [OLBC10, Chapter 10]. Using these formulas, we obtain, as real $z \to +\infty$,

$$\prod_{j=0}^{2} {}_0F_1(;\gamma_j;\omega^j z) \sim \frac{e^{\pi\sqrt{-1}(\gamma_2-\gamma_1)/3}\prod_{j=0}^{2}\Gamma(\gamma_j)}{8\pi^{3/2}}\, z^{3/4-(\gamma_0+\gamma_1+\gamma_2)/2}e^{4\sqrt{z}}. \tag{4.5}$$

Here each Bessel argument has argument $0, \pi/3, -\pi/3$, within the sector of that expansion. The displayed constant is nonzero for every $\gamma_j \notin \mathbb{Z}_{\le 0}$, so the product is not a polynomial. If all S_r were zero, (4.4) would make it a polynomial.

We choose a nonzero S_r. It has only finitely many zeros and poles, and $S_r(n+1)/S_r(n) \to 1$. The ratio $h_{n+1}(C_*)/h_n(C_*)$ is asymptotic to $(64/729)n^{-6}$. Multiplication by the nonterminating factors $(a)_{3n+r}(b)_{3n+r}$ gives a coefficient ratio tending to 64. Thus the series consisting of the terms whose degrees are congruent to r modulo 3 has radius of convergence $64^{-1/3} = 1/4$ in κ. The other residue classes occupy disjoint Taylor indices and cannot cancel it. Proposition 3.1 gives the lower bound $1/4$ for the radius of convergence of the full Taylor series. □

The same lower parameter C_* gives coefficient generation without factoring the numerator and denominator polynomials of S_r.

Corollary 4.5. *Under the assumptions of Corollary 4.4, we write the Taylor series of the restriction as $\sum_{N\ge 0} q_N \kappa^N$. We choose C_*, S_r, N_0 as in* (4.4) *and put*

$$\boldsymbol{A}_* = ((2C_*-1)/4, (2C_*+1)/4),$$
$$\boldsymbol{D}_* = (C_*, C_*/3, (C_*+1)/3, (C_*+2)/3, (2C_*-1)/3, 2C_*/3, (2C_*+1)/3).$$

For each $S_r \not\equiv 0$, the coefficients satisfy, at all sufficiently large indices,

(4.6)
$$\frac{q_{3n+3+r}}{q_{3n+r}} = 64\,\frac{\displaystyle\prod_{i=1}^{2}(n+A_{*,i})\prod_{j=0}^{2}\left(n+\frac{a+r+j}{3}\right)\prod_{j=0}^{2}\left(n+\frac{b+r+j}{3}\right)}{(n+1)\displaystyle\prod_{j=1}^{7}(n+D_{*,j})}\,\frac{S_r(n+1)}{S_r(n)}.$$

After the fixed rational functions and a finite initial segment are computed, the first N coefficients require $O(N)$ arithmetic operations.

Proof. We multiply (4.4) by $(a)_{3n+r}(b)_{3n+r}$ and use (2.7) at C_*. The resulting ratios give (4.6). Each nonzero S_r has finitely many integer zeros and poles, so one initial coefficient after these indices determines the whole tail. Residue classes with $S_r = 0$ have only the finite initial segment. Each step evaluates rational functions of fixed degrees, without finding their roots. This gives the arithmetic operation count. □

Remark 4.6. For the Lauricella differential system of [HT14, Theorem 11 and (2)], the singular locus outside the coordinate hyperplanes is given by

$$\mathcal{R}(x_1,x_2,x_3) = \prod_{\boldsymbol{\varepsilon}\in\{\pm1\}^3}(1+\varepsilon_1\sqrt{x_1}+\varepsilon_2\sqrt{x_2}+\varepsilon_3\sqrt{x_3}) = 0.$$

Here x_1, x_2, x_3 are the Lauricella variables, and the product is independent of the choices of square roots. We substitute $x_1 = \kappa, x_2 = \omega\kappa, x_3 = \omega^2\kappa$ and use square roots $t, \omega^2 t, \omega t$, where $t^2 = \kappa$. The eight signed sums are $0, 0, \pm 2t, \pm 2\omega t, \pm 2\omega^2 t$, so

$$\mathcal{R}(\kappa,\omega\kappa,\omega^2\kappa) = (1-4\kappa)(1-4\omega\kappa)(1-4\omega^2\kappa) = 1-64\kappa^3.$$

Thus the restricted curve avoids the singular locus outside the coordinate hyperplanes on $0 < |\kappa| < 1/4$. The original germ is holomorphic near zero and has trivial local monodromy there. Continuation of this germ therefore gives a single-valued holomorphic function on $|\kappa| < 1/4$, without assuming integer differences among the lower parameters. The finite ordinary ${}_8F_7$ reduction specifies this continuation explicitly when the lower parameters have integer differences. The Lauricella system has rank eight by [HT14, Theorem 12]. On an open subset of the restricted curve disjoint from the singular locus, its pullback gives a scalar differential equation of order at most eight for the restriction. The numbers of upper and lower parameters in (4.2) encode the rational functions R_r and do not determine a minimal differential order. By comparing coefficients in a fixed linear differential equation with polynomial coefficients, we obtain a recurrence with polynomial coefficients involving a fixed number of successive terms. After a finite initial segment, the nonzero leading polynomial of the recurrence has no integer zeros. Thus we can generate each coefficient with a fixed number of arithmetic operations. Thus the $O(N)$ count in Corollary 4.5 is also available from such an equation. When the lower parameters have integer differences, the construction gives an explicit first-order recurrence with rational functions of the index as coefficients, valid at all sufficiently large indices in each residue class whose coefficients are not eventually zero, without finding polynomial roots. Terminating upper parameters can give a radius larger than $1/4$.

We now determine the exact radius of convergence of the Taylor series without restricting the differences of the lower parameters.

Proposition 4.7. *Suppose that* $a, b, c_0, c_1, c_2 \notin \mathbb{Z}_{\leq 0}$. *Then the Taylor series at* $\kappa = 0$ *of*

$$F_C^{(3)}(a, b; c_0, c_1, c_2; \kappa, \omega\kappa, \omega^2\kappa)$$

has radius of convergence exactly $1/4$.

Proof. We denote the Taylor coefficients of the displayed restriction by q_N, and the coefficients of the entire product by u_N, so that

$$\prod_{j=0}^{2} {}_0F_1(;c_j;\omega^j z) = \sum_{N\geq 0} u_N z^N, \qquad q_N = (a)_N (b)_N u_N.$$

Coefficient comparison in the absolute convergence domain gives the second equality. Remark 4.6 gives the lower bound $1/4$. To prove the upper bound, we suppose that the radius of convergence R of the Taylor series is greater than $1/4$ and choose a real number ρ with $1/4 < \rho < R$. The Cauchy estimate gives $|q_N| \leq C\rho^{-N}$ for a constant $C > 0$. By the asymptotic formula for a quotient of gamma functions in [OLBC10, Eq. 5.11.12], we have

$$\frac{(a)_N (b)_N}{(N!)^2} \sim \frac{N^{a+b-2}}{\Gamma(a)\Gamma(b)}.$$

Since the upper parameters are nonterminating, for each $\eta > 1$ we can choose $C_\eta > 0$ such that

$$\frac{1}{|(a)_N (b)_N|} \leq C_\eta \frac{\eta^N}{(N!)^2} \qquad (N \geq 0).$$

We choose $1 < \eta < 4\rho$. For real $z > 0$, we obtain

$$\left| \prod_{j=0}^{2} {}_0F_1(;c_j;\omega^j z) \right| \leq CC_\eta \sum_{N\geq 0} \frac{(\eta z/\rho)^N}{(N!)^2} \leq CC_\eta e^{2\sqrt{\eta/\rho}\sqrt{z}}. \tag{4.7}$$

For the last inequality, we use $\binom{2N}{N} \leq 4^N$ and the even terms of the exponential series. The derivation of (4.5) applies whenever $c_j \notin \mathbb{Z}_{\leq 0}$ for $j = 0, 1, 2$, and gives

$$\prod_{j=0}^{2} {}_0F_1(;c_j;\omega^j z) \sim \frac{e^{\pi\sqrt{-1}(c_2-c_1)/3} \prod_{j=0}^{2} \Gamma(c_j)}{8\pi^{3/2}} z^{3/4-(c_0+c_1+c_2)/2} e^{4\sqrt{z}}.$$

The constant is nonzero, and the exponent $2\sqrt{\eta/\rho}$ in (4.7) is less than 4. This contradicts the displayed asymptotic and proves the upper bound. □

The proposition concerns only the radius of convergence of the Taylor series. The hypothesis that the lower parameters have integer differences remains in force for the finite reductions and the representation of the coefficients by the rational functions S_r at sufficiently large indices in (4.4).

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Graduate School of Science, Hokkaido University, Sapporo 060-0810, Japan
Email address: nakano.ryunosuke.i3@elms.hokudai.ac.jp